

Repository Inventory & Maintenance Log

Koch's Tree Nursery Tracker · Koch Clinical Suite · Pi Cabinet Monitor

Repository	kochmatthew462-collab/Bonsai-Grow-Nursery-Propigation
Head of main	7e52e50 2026-09-01 Merge pull request #12 from kochmatthew462-collab/claude/research-paper-generator-w6h7et
Commits (all branches)	43
Tracked files	111
Compiled	2026-09-06
Session	https://claude.ai/code/session_01Ju7ziEbYXfmFG4jX54a6Q1

Prepared by Matthew Koch. Parts 1–3 of the repository-management plan: export and index, permanent maintenance log, chat-based change control.

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Part 1 — Export & Index

Raw output of the export commands, run at the repository root on 2026-09-06. Only this repository was available in the session; the three *myomedprep.org* projects are not attached and their exports must be pasted separately.

[Project: Bonsai-Grow-Nursery-Propagation]

Git History

```
7e52e50 2026-09-01 kochmatthew462-collab Merge pull request #12 from kochmatthew462-collab/
claude/research-paper-generator-w6h7et
900eb73 2026-09-01 Claude Make the "already running" message describe the run that is running
7cc36fe 2026-09-01 kochmatthew462-collab Merge pull request #11 from kochmatthew462-collab/
claude/research-paper-generator-w6h7et
7f1eeb1 2026-09-01 Claude Let the suite be reached from another machine, deliberately
365de62 2026-09-01 kochmatthew462-collab Merge pull request #7 from kochmatthew462-collab/claude/
research-paper-generator-w6h7et
a257bec 2026-08-30 kochmatthew462-collab Merge pull request #9 from kochmatthew462-collab/claude/
bonsai-metrics-tracking-h8r68l
65f59fb 2026-08-30 Claude Fix Pi install for Bookworm (PEP 668) and non-'pi' user accounts
8652b77 2026-08-29 Claude Merge main: keep both .gitignore additions
00c8563 2026-08-28 kochmatthew462-collab Merge pull request #8 from kochmatthew462-collab/claude/
bonsai-metrics-tracking-h8r68l
5514c08 2026-08-29 Claude Add Raspberry Pi cabinet monitor: continuous sensors, live card, chill
logger
026ef9f 2026-08-19 Claude Stop making the user re-authenticate every time they close the tab
5b03b56 2026-08-19 Claude Stop the port from moving, which is what was 404ing
530ff72 2026-08-19 Claude Put APA 7 first, and make it readable without a project
1d73ea0 2026-08-19 Claude Give APA 7 its own screen
f2044d5 2026-08-19 Claude Forget a rejected token, and let one be pasted in
3dd82c6 2026-08-19 Claude Keep the session token across restarts
48c0f1a 2026-08-19 Claude Stop the browser caching the shell and its scripts
ce20031 2026-08-19 Claude Explain a stale token instead of throwing on it
2191825 2026-08-19 Claude Declare both ports and open the browser automatically
17bd465 2026-08-19 Claude Stop recommending a command that is not installed
d91b8d2 2026-08-18 Claude Let the self check run outside the virtual environment
4bcc8e0 2026-08-18 Claude Add a self check, because four rounds of guessing is three too many
071f537 2026-08-18 Claude Bind wide in a Codespace, and declare the forwarded port
b826cdf 2026-08-18 Claude Make it work in a GitHub Codespace, without loosening the host check
72e4def 2026-08-18 Claude Keep the nav bars, and hide only the steps that need something open
589fdad 2026-08-17 Claude Make the launcher say why it will not start
de6bb89 2026-08-17 Claude Never print a URL that nothing is listening on
bc86193 2026-08-17 Claude Fix a blank screen caused by CSS defeating the hidden attribute
11c4438 2026-08-17 Claude Read PDFs into anchored passages, and wire up what was unreachable
9094033 2026-08-17 Claude Fix defects found by an adversarial audit of this codebase
18d24b7 2026-08-17 Claude Add discovery, compliance and statistics tooling
9b4000a 2026-08-17 Claude Fix a syntax error that broke the entire UI, and test for it
074715f 2026-08-17 Claude Add spelling and grammar checking to the charting tab
4cfab32 2026-08-17 Claude Add clinical charting tab for bedside and advanced-practice nurses
fd76469 2026-08-17 Claude Add evidence-based research and APA 7 authoring suite
32ded12 2026-08-17 kochmatthew462-collab Add professional record layer: specimen, lineage, IPM,
photos, derived readouts (#6)
39ffeb7 2026-08-16 kochmatthew462-collab Add weather watch, move alerts, and calendar export (#5)
3d97ca5 2026-08-16 kochmatthew462-collab Add master calendar, derived drip log, and handbook task
windows (#4)
ab26e88 2026-08-16 kochmatthew462-collab Align care profiles with the three handbooks; rename to
Koch's Tree Nursery Tracker (#3)
```

9f8f639	2026-08-16	kochmatthew462-collab	Add environment factors, care profiles, target bands and optional cloud sync (#2)
f78c115	2026-08-16	kochmatthew462-collab	Add .nojekyll so Pages publishes the static files as-is
750d52c	2026-08-16	kochmatthew462-collab	Add bonsai nursery tracker with scannable QR plant labels (#1)
5569c01	2026-08-16	kochmatthew462-collab	Initial commit

Remotes

```
origin https://github.com/kochmatthew462-collab/Bonsai-Grow-Nursery-Propigation (fetch)
origin https://github.com/kochmatthew462-collab/Bonsai-Grow-Nursery-Propigation (push)
```

Branch Structure

```
* claude/repos-pdf-maintenance-log-jlwmtl
  main
  remotes/origin/claude/repos-pdf-maintenance-log-jlwmtl
  remotes/origin/main
```

File Inventory

```
.devcontainer/devcontainer.json
.gitattributes
.gitignore
.nojekyll
README.md
css/styles.css
index.html
js/app.js
js/calendar.js
js/charts.js
js/photos.js
js/profiles.js
js/qrcode.js
js/store.js
js/sync.js
js/weather.js
pi/README.md
pi/cloud.py
pi/config.example.json
pi/install.sh
pi/plantmon.service
pi/sensord.py
pi/sensors.py
pi/store.py
pi/test_pi.py
research-suite/.env.example
research-suite/README.md
research-suite/app/__init__.py
research-suite/app/apa/__init__.py
research-suite/app/apa/assemble.py
research-suite/app/apa/audit_document.py
research-suite/app/apa/citations.py
research-suite/app/apa/deck.py
research-suite/app/apa/document.py
research-suite/app/apa/figures.py
research-suite/app/apa/ooxml.py
research-suite/app/apa/prisma.py
research-suite/app/apa/workbook.py
```

research-suite/app/charting/__init__.py
research-suite/app/charting/disclosure.py
research-suite/app/charting/export.py
research-suite/app/charting/interlocks.py
research-suite/app/charting/language.py
research-suite/app/charting/models.py
research-suite/app/charting/narrative.py
research-suite/app/charting/phi.py
research-suite/app/charting/proofing.py
research-suite/app/charting/routes.py
research-suite/app/charting/scales.py
research-suite/app/charting/specialty.py
research-suite/app/charting/store.py
research-suite/app/charting/systems.py
research-suite/app/compliance/__init__.py
research-suite/app/compliance/journals.py
research-suite/app/compliance/rubric.py
research-suite/app/compliance/simulator.py
research-suite/app/doctor.py
research-suite/app/evidence/__init__.py
research-suite/app/evidence/appraisal.py
research-suite/app/evidence/dedupe.py
research-suite/app/evidence/levels.py
research-suite/app/main.py
research-suite/app/models.py
research-suite/app/research/__init__.py
research-suite/app/research/pico.py
research-suite/app/security.py
research-suite/app/settings.py
research-suite/app/sources/__init__.py
research-suite/app/sources/base.py
research-suite/app/sources/fulltext.py
research-suite/app/sources/gov.py
research-suite/app/sources/importers.py
research-suite/app/sources/scholarly.py
research-suite/app/static/app.js
research-suite/app/static/chart.js
research-suite/app/static/index.html
research-suite/app/static/shell.js
research-suite/app/static/styles.css
research-suite/app/storage.py
research-suite/app/writing/__init__.py
research-suite/app/writing/draft.py
research-suite/app/writing/integrity.py
research-suite/app/writing/proof.py
research-suite/app/writing/statistics.py
research-suite/app/writing/style.py
research-suite/requirements.txt
research-suite/run.ps1
research-suite/run.sh
research-suite/tests/__init__.py
research-suite/tests/run_all.sh
research-suite/tests/test_apa_citations.py
research-suite/tests/test_charting_flow.py
research-suite/tests/test_charting_language.py
research-suite/tests/test_charting_proofing.py
research-suite/tests/test_charting_scales.py
research-suite/tests/test_deck.py
research-suite/tests/test_docx_layout.py
research-suite/tests/test_evidence.py
research-suite/tests/test_frontend.py
research-suite/tests/test_fulltext.py
research-suite/tests/test_research_tools.py
research-suite/tests/test_security.py
research-suite/tests/test_sources.py

```
test/calendar_test.py
test/emit_matrices.js
test/pro_test.py
test/sensors_test.py
test/smoke_app.py
test/sync_test.py
test/verify_qr.py
test/weather_test.py

--- ls -lah . ---
total 92K
drwxr-xr-x 10 root root 4.0K Sep  6 03:16 .
drwxr-xr-x  3 root root 4.0K Sep  6 03:15 ..
drwxr-xr-x  2 root root 4.0K Sep  6 03:15 .devcontainer
drwxr-xr-x  8 root root 4.0K Sep  6 03:15 .git
-rw-r--r--  1 root root  986 Sep  6 03:15 .gitattributes
-rw-r--r--  1 root root  698 Sep  6 03:15 .gitignore
-rw-r--r--  1 root root  276 Sep  6 03:15 .nojekyll
-rw-r--r--  1 root root  33K Sep  6 03:15 README.md
drwxr-xr-x  2 root root 4.0K Sep  6 03:15 css
drwxr-xr-x  3 root root 4.0K Sep  6 03:16 docs
-rw-r--r--  1 root root 2.4K Sep  6 03:15 index.html
drwxr-xr-x  2 root root 4.0K Sep  6 03:15 js
drwxr-xr-x  2 root root 4.0K Sep  6 03:15 pi
drwxr-xr-x  4 root root 4.0K Sep  6 03:15 research-suite
drwxr-xr-x  2 root root 4.0K Sep  6 03:15 test
```

Domain 11 — Code Repos

11.1 Repository record

Field	Value
Name	Bonsai-Grow-Nursery-Propagation
Owner	kochmatthew462-collab
URL	https://github.com/kochmatthew462-collab/Bonsai-Grow-Nursery-Propagation
Clone (HTTPS)	https://github.com/kochmatthew462-collab/Bonsai-Grow-Nursery-Propagation.git
Hosting	GitHub (cloud). No self-hosted mirror.
CI/CD	None. No .github/workflows. Tests are run by hand before pushing.
Deploy: tracker	GitHub Pages, deploy-from-branch, root folder, .nojekyll present
Deploy: clinical suite	Not deployed. research-suite/run.sh locally, or Codespace port 8765 (private)
Deploy: Pi monitor	pi/install.sh on the Pi → systemd unit plantmon
Default branch	main — head 7e52e50, 2026-09-01
Contributors	kochmatthew462-collab (13 commits), Claude via Claude Code sessions (30 commits)
Size	111 tracked files, ~2.5 MB working tree, ~43,600 lines of source and tests
Secrets policy	.env, research-suite/data/, .session-token, pi/config.json, *.db git-ignored; committed key = rotate

11.2 Branches and their purposes

Branch	On remote	Purpose	Last commit	State
main	yes	Default; what GitHub Pages publishes	7e52e50 2026-09-01 Merge PR #12	current
claude/research-paper-generator-w6h7et	yes (via PR #13 head 479490f)	Claude Code session: clinical suite, APA 7, charting, Codespace/token fixes	479490f	PR #13 open
claude/bonsai-metrics-tracking-h8r68l	no (deleted after merge)	Claude Code session: tracker, profiles, calendar, weather, Pi monitor	65f59fb 2026-08-30	fully merged via PR #9

Branch	On remote	Purpose	Last commit	State
claude/radiologic-imaging-interpreter-tlr236	no	Claude Code session: Radiology Interpretation Academy	6c533ac	PR #10 closed, not merged
claude/repos-pdf-maintenance-log-jlwmtl	yes	This inventory, MAINTENANCE_LOG.md , this PDF	see PR	open

11.3 Pull request ledger

PR	Title	Session branch	Merged
#1	Add bonsai nursery tracker with scannable QR plant labels	bonsai-metrics-tracking	2026-08-16 (squash 750d52c)
#2	Add environment factors, care profiles, target bands and optional cloud sync	bonsai-metrics-tracking	2026-08-16 (squash 9f8f639)
#3	Align care profiles with the three handbooks; rename to Koch's Tree Nursery Tracker	bonsai-metrics-tracking	2026-08-16 (ab26e88)
#4	Add master calendar, derived drip log, and handbook task windows	bonsai-metrics-tracking	2026-08-17 (3d97ca5)
#5	Add weather watch, move alerts, and calendar export	bonsai-metrics-tracking	2026-08-17 (39ffeb7)
#6	Add professional record layer: specimen, lineage, IPM, photos, derived readouts	bonsai-metrics-tracking	2026-08-17 (32ded12)
#7	Clinical suite: APA 7 research authoring with source-anchored claims, plus a nursing charting tab	research-paper-generator	2026-09-01 (merge 365de62 , 24 commits)
#8	Add Raspberry Pi cabinet monitor: continuous sensors, live card, chill logger	bonsai-metrics-tracking	2026-08-29 (00c8563)
#9	Fix Pi install for Bookworm (PEP 668) and non-'pi' user accounts	bonsai-metrics-tracking	2026-08-30 (a257bec)
#10	Add Radiology Interpretation Academy — self-study radiologic imaging interpretation app	radiologic-imaging-interpreter	closed, not merged
#11	Let the suite be reached from another machine, deliberately	research-paper-generator	2026-09-01 (7cc36fe)
#12	Make the "already running" message describe the run that is running	research-paper-generator	2026-09-01 (7e52e50)
#13	Add a place to actually write in APA 7	research-paper-generator	open

11.4 File structure

index.html	tracker shell and routes
css/styles.css	theme tokens, light and dark
js/	app.js · store.js · profiles.js · calendar.js weather.js · charts.js · photos.js · sync.js · qrcode.js
test/	tracker tests (7 scripts + emit_matrices.js)
pi/	sensord.py · sensors.py · store.py · cloud.py install.sh · plantmon.service · config.example.json test_pi.py · README.md
research-suite/	
run.sh / run.ps1	launcher; creates and refreshes .venv
requirements.txt	fastapi, uvicorn, python-docx, python-pptx, matplotlib, anthropic, keyring, pypdf...
.env.example	every key optional; HOST/PORT/ALLOWED_HOSTS
app/	
main.py · security.py · settings.py · storage.py	
models.py · doctor.py	
apa/	citations, document, ooxml, assemble, audit_document, figures, prisma, workbook, deck
charting/	routes, models, store, interlocks, scales, language, proofing, phi, narrative, export...
compliance/	rubric, simulator, journals
evidence/	levels, appraisal, dedupe
research/	pico.py
sources/	base, scholarly, gov, fulltext, importers
writing/	draft, style, proof, integrity, statistics
static/	index.html · shell.js · app.js · chart.js · styles.css
tests/	run_all.sh + 13 offline suites
docs/	
repo-inventory/	raw git and file exports (Part 1)
build_repo_pdf.py	rebuilds this PDF
Repo-Inventory-and-Maintenance-Log.pdf	
MAINTENANCE_LOG.md	source of truth for decisions
.devcontainer/	Codespace: port 8765 private, auto-start
.gitattributes	LF for *.sh; binaries marked
.gitignore	secrets, data, venv, caches
.nojekyll	Pages publishes as-is
README.md	tracker documentation (33 KB)

11.5 Which Claude Code sessions touched which files

Session	Files
bonsai-metrics-tracking-h8r68l (PRs #1–#6)	index.html, css/styles.css, js/app.js, js/store.js, js/profiles.js, js/calendar.js, js/weather.js, js/charts.js, js/photos.js, js/sync.js, js/qrcode.js, test/*, README.md, .nojekyll
bonsai-metrics-tracking-h8r68l (PR #8)	pi/* (all), js/app.js, js/sync.js, test/sensors_test.py, README.md, .gitignore
bonsai-metrics-tracking-h8r68l (PR #9)	pi/README.md, pi/install.sh, pi/plantmon.service

Session	Files
research-paper-generator-w6h7et (PR #7)	research-suite/** (all 80 files), .devcontainer/devcontainer.json, .gitattributes, .gitignore
research-paper-generator-w6h7et (PR #11)	research-suite/app/{main,security,settings,doctor}.py, app/static/app.js, tests/test_security.py, README.md
research-paper-generator-w6h7et (PR #12)	research-suite/app/{main,security}.py, tests/test_security.py, README.md
repos-pdf-maintenance-log-jlwmtl (this)	MAINTENANCE_LOG.md, docs/**

Koch's Tree Nursery Tracker & Koch Clinical Suite

— Maintenance & Architecture Log

Overview

- **Repo:** <https://github.com/kochmatthew462-collab/Bonsai-Grow-Nursery-Propagation>
- **Default branch:** `main`
- **Live:**
- Nursery tracker — GitHub Pages, published from the repo root of the chosen branch (Settings → Pages → deploy from a branch → `/` (root)). Static; `.nojekyll` keeps Pages from running Jekyll.
- Clinical suite (`research-suite/`) — **not hosted**. Runs locally at `127.0.0.1:8765`, or in a GitHub Codespace on forwarded port 8765 (private visibility).
- Pi cabinet monitor (`pi/`) — installed on a Raspberry Pi 3B+ under systemd as `plantmon`; writes to the same Firestore document the app syncs.
- **Last updated:** 2026-09-06
- **Last commit on** `main`: `7e52e50` (2026-09-01) — Merge pull request #12
- **Raw inventory:** `docs/repo-inventory/` (git log, remotes, branches, file tree, files touched per session)
- **Current issues/debt:** see [Known Issues & Debt](#)

This log is the source of truth for architectural decisions. Every decision or problem gets an entry here **with a link to the session or pull request that made it**. Claude Code sessions are archive-only context; the log is what is maintained.

Three products, one repository

Product	Path	Stack	Runs where	Tests
Koch's Tree Nursery Tracker	<code>index.html</code> , <code>css/</code> , <code>js/</code>	Static HTML/CSS/JS, no build, no CDN	GitHub Pages, any browser, <code>file://</code>	<code>test/*.py</code> (Playwright), <code>test/verify_qr.py</code>
Koch Clinical Suite (research & APA 7 authoring + clinical charting)	<code>research-suite/</code>	Python 3.11+, FastAPI, uvicorn, python-docx, python-pptx, matplotlib	Local machine or Codespace, loopback only	<code>research-suite/tests/run_all.sh</code> (13 suites, offline)
Cabinet monitor	<code>pi/</code>	Python 3 on Raspberry Pi OS, smbus2, SQLite, Firestore REST	Raspberry Pi 3B+, systemd	<code>pi/test_pi.py</code>

Architecture Decision Record (ADR)

ADR-001: Static site, no server, no accounts (tracker)

- **Date:** 2026-08-16
- **Decision:** The tracker is plain static files served by GitHub Pages. Data lives in the browser (localStorage) with optional sync. No build step, no bundler, no CDN dependency.
- **Rationale:** Labels must print and scan from a phone in a greenhouse with no signal and from a `file:///` copy. [PR #1](#)
- **Status:** LOCKED
- **Impact:** Every feature must work offline; nothing secret may live in this half of the repo (see ADR-006).

ADR-002: QR codes, not 1D barcodes; encoder written from scratch

- **Date:** 2026-08-16
- **Decision:** Every label is a QR code carrying the plant's full URL. `js/qrcode.js` is a self-contained encoder (versions 1–10, EC levels L and M, URLs up to 213 bytes).
- **Rationale:** A phone camera opens a URL with nothing installed; a 1D barcode needs a scanner app plus a lookup. A CDN encoder fails offline. [PR #1](#)
- **Status:** LOCKED
- **Impact:** `test/verify_qr.py` decodes 138 matrices through `zxing-cpp` and compares function patterns to `segno`. Any change to the encoder must pass it.

ADR-003: Optional Firestore sync over REST, merge-by-record, tombstones

- **Date:** 2026-08-16
- **Decision:** Cross-device sync uses the Cloud Firestore REST API via `fetch` with anonymous auth. No Firebase SDK. Each plant and each check merges independently, newest edit wins; deletes are tombstones. The nursery code (24 chars, ~120 bits) is the only credential.
- **Rationale:** Free Spark plan, does not pause when idle, keeps the app a set of plain files. [PR #2](#)
- **Status:** LOCKED
- **Impact:** Shared document has a 1 MiB ceiling; app warns at 800 KB. Sync is not backup — JSON export remains the recovery path. The Pi (ADR-005) must obey the same merge rules.

ADR-004: Care profiles aligned to the three handbooks

- **Date:** 2026-08-16
- **Decision:** Tracked factors, target bands, seasonal windows and the master calendar are derived from the indoor bench, bergamot and olive handbooks, and the app was renamed Koch's Tree Nursery Tracker.
- **Rationale:** [PR #3](#), [PR #4](#), [PR #5](#)
- **Status:** LOCKED

- **Impact:** `js/profiles.js` is the single place bands live. README records where the app's dates deliberately differ from the handbooks.

ADR-005: The Raspberry Pi is "one more device on the nursery"

- **Date:** 2026-08-29
- **Decision:** The Pi daemon uses the same three Sync credentials and the same merge rules as the app (ported to Python, proven byte-identical against `js/store.js` in `pi/test_pi.py`). Raw one-minute readings stay in SQLite on the Pi; a live document every 5 minutes feeds the Live sensors card; one minimal summary entry per plant per day merges under a deterministic id. Relays are low-voltage only and disabled by default.
- **Rationale:** No port forwarding, no new server, nothing new to secure. [PR #8](#), [PR #9](#)
- **Status:** LOCKED
- **Impact:** A year of summaries for six plants is ~250 KB of the 1 MiB document. Stale after 15 quiet minutes is a loud state, by design.

ADR-006: The clinical suite is a local app, never a static site

- **Date:** 2026-08-17
- **Decision:** `research-suite/` binds to `127.0.0.1`, requires a session token, checks the `Host` header against an allowlist taken from the environment (never from the request), and checks `Origin` on writes. API keys live in the OS keychain or a git-ignored `.env`.
- **Rationale:** It holds API keys and calls APIs with no CORS headers; a key in front-end JavaScript is a published key. Because this repo publishes GitHub Pages from its own branch, a committed key would be served at a guessable URL. [PR #7](#)
- **Status:** LOCKED
- **Impact:** `.env`, `data/`, `.session-token` are git-ignored. A key that has ever been committed must be rotated. Binding wide (`RESEARCH_SUITE_HOST=0.0.0.0`) does not widen admission; hosts must be named in `RESEARCH_SUITE_ALLOWED_HOSTS` ([PR #11](#)).

ADR-007: Session token is stable and carried in an HttpOnly cookie

- **Date:** 2026-08-19
- **Decision:** The token is generated once into `data/.session-token` (0600), not per run. On first visit it is written to an `HttpOnly`, `SameSite=Lax` cookie (Secure over HTTPS, 30 days). A rejected cookie is deleted by the rejection. Rotate by deleting the file; pin with `RESEARCH_SUITE_TOKEN`.
- **Rationale:** Regenerating per run invalidated every open tab on restart; `sessionStorage` lost it on tab close. Commits `3dd82c6`, `026ef9f`, `ce20031`, `f2044d5` in [PR #7](#).
- **Status:** LOCKED
- **Impact:** Bookmark the plain address without `#token=`. The real boundary is the OS account plus the Host allowlist, not the token.

ADR-008: Fixed port 8765; Codespace starts the app on container start

- **Date:** 2026-08-18 / 2026-08-19
- **Decision:** One port, 8765, declared in `.devcontainer/devcontainer.json` with private visibility and `onAutoForward: openBrowser`. `postStartCommand` launches `run.sh` detached, logging to `/tmp/koch-suite.log`. `python -m app.doctor` diagnoses the layers when the forwarded URL 404s.
- **Rationale:** A moving port and an un-forwarded port both look like a broken app. Commits `5b03b56`, `2191825`, `4bcc8e0` in [PR #7](#).
- **Status:** LOCKED
- **Impact:** Devcontainer changes take effect on **rebuild**, not restart.

ADR-009: Drafting produces a claim ledger, not prose; no AI-detection gate; no "humanizer"

- **Date:** 2026-08-17
- **Decision:** The research half anchors every claim to a source, page and paragraph. No plagiarism percentage is printed; unattributed verbatim overlap against cited sources is reported with offsets. No AI-detector gate and no detector-evasion feature. Stylometric calibration against the author's own samples is provided instead.
- **Rationale:** Detectors measure perplexity, not authorship; evasion tools are unstable and presume the work is not the author's. Documented in `research-suite/README.md`, [PR #7](#).
- **Status:** LOCKED
- **Impact:** Any request to add a similarity percentage, an AI-detection gate, or paraphrase-to-evade is refused by design.

ADR-010: Charting holds no PHI; instruments re-implemented, never reproduced

- **Date:** 2026-08-17
- **Decision:** No field for name, MRN or DOB; encounters keyed by bed/room; free text scanned for identifiers with one-click redaction; purge on every screen. Scoring arithmetic and thresholds for 23 instruments are implemented with fresh item wording; no copyrighted text or artwork (e.g. Wong-Baker FACES) is reproduced; rights holders are named on the Reference screen. ESI is walked as decision points the user answers, never auto-suggested.
- **Rationale:** HIPAA exposure on a personal device lands on the nurse; a paraphrased near-copy of an instrument is worse than either extreme; auto-triage is clinical decision support and indefensible in deposition. [PR #7](#)
- **Status:** LOCKED
- **Impact:** `test_charting_scales.py` asserts every instrument names its rights holder; `test_charting_language.py` holds the negative assertions.

ADR-011: Every test suite runs offline

- **Date:** 2026-08-17

- **Decision:** No test needs network, keys or credentials. Retrieval adapters are tested against `httpx.MockTransport`; sync against a mocked Firestore; the front end with `node --check` plus orphan-route detection.
- **Rationale:** A syntax error once shipped that took out the whole UI and no Python test noticed (9b4000a). [PR #7](#)
- **Status:** LOCKED
- **Impact:** `bash research-suite/tests/run_all.sh` is the pre-push gate for the suite.

ADR-012: Repository management moves to the chat-based maintenance log

- **Date:** 2026-09-06
- **Decision:** New work is decided and logged here first, with a session link, then applied to the repo. Claude Code session branches become archive-only context. This file plus `docs/repo-inventory/` is the "Domain 11: Code Repos" record.
- **Rationale:** Decisions spread across a dozen session branches were not searchable. [Session](#)
- **Status:** OPEN — awaiting the owner's confirmation of the deployment method (see Immediate Action Items)
- **Impact:** Each future PR description should cite the ADR or issue row it resolves.

Known Issues & Debt

Issue	Impact	Session / PR	Status
PR #13 "Add a place to actually write in APA 7" is open and unmerged (head 479490f on <code>claude/research-paper-generator-w6h7et</code>)	research suite feature gap	PR #13	needs review and merge decision
PR #10 Radiology Interpretation Academy was closed without merging; branch <code>claude/radiologic-imaging-interpreter-tlr236</code> is not on the remote	work possibly lost; scope question (does it belong in this repo?)	PR #10	needs owner decision
Three products share one repo and one Pages deployment; <code>research-suite/</code> source is published as static files by Pages	any committed secret is served publicly	ADR-006	mitigated by <code>.gitignore</code> ; consider splitting repos
Firestore shared document has a 1 MiB hard ceiling; sensor summaries accrue ~250 KB/yr for six plants	sync stops working when exceeded	ADR-003, ADR-005	monitor the 800 KB warning; prune yearly

Issue	Impact	Session / PR	Status
No CI/CD: no <code>.github/workflows</code> ; tests run only by hand	regressions reach <code>main</code> unnoticed	—	open: add a GitHub Actions job running <code>research-suite/tests/run_all.sh</code> and <code>node --check js/*.js</code>
Tracker tests need <code>segno</code> <code>zxing-cpp</code> <code>numpy</code> <code>playwright</code> <code>pillow</code> installed manually	friction	README "Tests"	open: add a <code>requirements-test.txt</code>
Gated databases (CINAHL, PsycINFO, Embase, Cochrane, JBI, Scopus, IEEE, JSTOR) cannot be queried by API	manual citation-file import is the only path	<code>research-suite/README.md</code>	by design, will not change
Grammarly cannot be integrated; LanguageTool must be self-hosted via Docker	grammar checking degrades silently to off when server absent	<code>research-suite/README.md</code>	by design
Pi wiring cannot be proven by tests; <code>i2cdetect</code> and <code>--once --no-cloud</code> are the bench checks	a probe pulled loose reads dry forever	<code>pi/README.md</code>	operational, documented
Codespace data (keys, projects) lives only in the container	lost on container deletion	ADR-008	export regularly

How to Deploy

```
# Nursery tracker (GitHub Pages) – nothing to build
git checkout main
git pull origin main
# edit index.html / css / js
python3 -m http.server 8000          # local check at http://localhost:8000
git commit -am "..." && git push origin main
# Pages re-publishes main automatically (Settings → Pages: deploy from a branch,
main, / (root))

# Clinical suite – local machine
cd research-suite
./run.sh                            # or ./run.ps1 on Windows; makes .venv on first run
python3 -m app.doctor --url          # prints the launch URL with the token
# In a Codespace: port 8765 forwards itself and opens the browser; rebuild the
# container after changing .devcontainer/devcontainer.json.

# Clinical suite – always-on box on the LAN
RESEARCH_SUITE_HOST=0.0.0.0 \
RESEARCH_SUITE_ALLOWED_HOSTS=pi-3bplus.local,192.168.1.50 \
./run.sh

# Pi cabinet monitor
# on the Pi, inside pi/ (clone the repo or copy pi/ over)
bash install.sh
nano ~/plantmon/config.json          # Sync credentials, plant ids, probe calibration
```

```
python3 ~/plantmon/sensord.py --config ~/plantmon/config.json --once --no-cloud
sudo systemctl start plantmon
journalctl -u plantmon -f
```

How to Test (run before every push)

```
# Clinical suite – all 13 suites, offline
bash research-suite/tests/run_all.sh

# Tracker
pip install segno zxing-cpp numpy playwright pillow
python3 test/verify_qr.py
python3 test/smoke_app.py
python3 test/sync_test.py
python3 test/calendar_test.py
python3 test/weather_test.py
python3 test/pro_test.py
python3 test/sensors_test.py
python3 pi/test_pi.py
```

Debugging & Common Problems

"Forwarded Codespace URL returns 404"

Solution: ADR-008 — nothing is listening. Run `python3 -m app.doctor`; it says whether the app is not running, on another port, bound to loopback only, or the port was never forwarded. If the devcontainer was edited, rebuild the container.

"Missing or invalid session token" on every reload

Solution: ADR-007 — the cookie was rejected and deleted. Get the launch URL back with `python3 -m app.doctor --url`, open it once, then bookmark the address without `#token=`.

"421 Misdirected Request" from another machine

Solution: PR #11 — binding wide is not admission. Add the address you are browsing from to `RESEARCH_SUITE_ALLOWED_HOSTS`.

"bash: ./run.sh: /usr/bin/env bash^M: bad interpreter"

Solution: CRLF checkout on Windows. `.gitattributes` now pins `*.sh` to LF; re-clone or run `git checkout -- run.sh` after `git config core.autocrlf false`. `bash run.sh` also works.

"Suite starts but Word export is missing"

Solution: `requirements.txt` has no optional split on purpose. `./run.sh` refreshes the venv when requirements change; delete `research-suite/.venv` and rerun if a wheel failed.

"Blank screen in the suite"

Solution: commits `bc86193` and `9b4000a` — a CSS rule defeated the `hidden` attribute, and a syntax error in `app.js`. `python3 research-suite/tests/test_frontend.py` runs `node --check` on every script and catches both.

"Live sensors card says stale"

Solution: ADR-005 — the Pi has not written the live document for 15 minutes. On the Pi: `journalctl -u plantmon -f`, then `i2cdetect -y 1` (expect 29, 5c, 68, 70, 48 and 49/4a/4b). A missing address is wiring.

"Scanned a label and got an empty plant"

Solution: README "Sync" — this device has never seen the plant. Join the nursery with the same code on the Sync page or import the JSON backup.

"Sync stopped; document too large"

Solution: ADR-003 — export a JSON backup and prune old sensor entries; the shared document ceiling is 1 MiB.

"Pi install fails: externally-managed-environment"

Solution: PR #9 — Bookworm refuses system-wide pip. `install.sh` now installs `python3-smbus2` and `python3-requests` from apt.

Session Index (which sessions touched which files)

Session branch	PRs	Dates	Files touched
<code>claude/bonsai-metrics-tracking-h8r68l</code>	#1–#6 (squash-merged), #8, #9	2026-08-16 → 2026-08-30	<code>index.html</code> , <code>css/styles.css</code> , <code>js/*.js</code> , <code>test/*</code> , <code>pi/*</code> , <code>README.md</code> , <code>.gitignore</code>
<code>claude/research-paper-generator-w6h7et</code>	#7, #11, #12 merged; #13 open	2026-08-17 → 2026-09-01	<code>research-suite/**</code> , <code>.devcontainer/devcontainer.json</code> , <code>.gitattributes</code> , <code>.gitignore</code>
<code>claude/radiologic-imaging-interpreter-tlr236</code>	#10 closed, not merged	2026-08-31	none on <code>main</code> ; branch absent from remote
<code>claude/repos-pdf-maintenance-log-jlwmtl</code>	this log	2026-09-06	<code>MAINTENANCE_LOG.md</code> , <code>docs/repo-inventory/*</code> , <code>docs/*.pdf</code>

Full per-merge file lists: `docs/repo-inventory/files_by_session.txt`.

Immediate Action Items

1. **Confirm deployment method** for the tracker: which branch does GitHub Pages currently publish, and what is the published URL? Record it in Overview.
 2. **Decide PR #13** (merge, revise, or close) and **PR #10** (recover the radiology app into its own repo, or abandon).
 3. **Add CI**: one GitHub Actions workflow running the suite tests and `node --check`.
 4. **Lock this log**: from here on, every change starts as an ADR or issue row with a session link, then a PR that cites it.
-

Change log for this file

Date	Change	Session
2026-09-06	Created from the full repo inventory; ADR-001 to ADR-012 recorded	session_01Ju7ziEbYXfmFG4jX54a6Q1

Part 3 — Chat-based change control

From 2026-09-06 (ADR-012) repository work follows this loop:

- 1. **Ask in the maintenance chat**, not in an ad-hoc Claude Code session.
- 2. The chat produces one of: code to paste into the local repo and push; git commands to export or import from the live target; or a deployment script to run from the terminal.
- 3. **Every decision is logged** in `MAINTENANCE_LOG.md` as an ADR or an issue row, with the session link, before the PR is opened. The PR description cites the ADR or row it resolves.
- 4. Claude Code branches are archive-only: consult them for old context; do not build new work on them.

Entry template

Session [link]: Decided *what*, because *why*. Updated `MAINTENANCE_LOG.md` ADR-NNN to *STATUS*. Deployment plan: *steps*, *target*, *verification*.

Deployment facts still to confirm

Question	Current answer from the repo	Owner to confirm
Are these repos on GitHub/GitLab?	GitHub, <code>kochmatthew462-collab/Bonsai-Grow-Nursery-Propagation</code>	✓
Self-hosted?	No. Pages for the tracker; local/Codespace for the suite; a Pi on the LAN for the monitor	—
CI/CD pipeline?	None	decide whether to add GitHub Actions
How are changes deployed today?	Merge to <code>main</code> → Pages republishes; suite and Pi are pulled by hand	confirm which branch Pages publishes and its URL

Locking the schema

Once the table above is confirmed, ADR-012 moves to `LOCKED` and `MAINTENANCE_LOG.md` becomes the sole source of truth for architecture. The exports in `docs/repo-inventory/` are a snapshot; regenerate them with the commands in Part 1 and rebuild this PDF with `python3 docs/build_repo_pdf.py` whenever the inventory changes.

Appendix — Files touched per session merge

```
== merge a257bec: Merge pull request #9 from kochmatthew462-collab/claude/bonsai-metrics-tracking-
h8r68l
  pi/README.md
  pi/install.sh
  pi/plantmon.service
== merge 00c8563: Merge pull request #8 from kochmatthew462-collab/claude/bonsai-metrics-tracking-
h8r68l
  .gitignore
  README.md
  js/app.js
  js/sync.js
  pi/README.md
  pi/cloud.py
  pi/config.example.json
  pi/install.sh
  pi/plantmon.service
  pi/sensord.py
  pi/sensors.py
  pi/store.py
  pi/test_pi.py
  test/sensors_test.py
== merge 7e52e50: Merge pull request #12 from kochmatthew462-collab/claude/research-paper-generator-
w6h7et
  research-suite/README.md
  research-suite/app/main.py
  research-suite/app/security.py
  research-suite/tests/test_security.py
== merge 7cc36fe: Merge pull request #11 from kochmatthew462-collab/claude/research-paper-generator-
w6h7et
  research-suite/README.md
  research-suite/app/doctor.py
  research-suite/app/main.py
  research-suite/app/security.py
  research-suite/app/settings.py
  research-suite/app/static/app.js
  research-suite/tests/test_security.py
== merge 365de62: Merge pull request #7 from kochmatthew462-collab/claude/research-paper-generator-
w6h7et
  .devcontainer/devcontainer.json
  .gitattributes
  .gitignore
  research-suite/.env.example
  research-suite/README.md
  research-suite/app/__init__.py
  research-suite/app/apa/__init__.py
  research-suite/app/apa/assemble.py
  research-suite/app/apa/audit_document.py
  research-suite/app/apa/citations.py
  research-suite/app/apa/deck.py
  research-suite/app/apa/document.py
  research-suite/app/apa/figures.py
  research-suite/app/apa/ooxml.py
  research-suite/app/apa/prisma.py
  research-suite/app/apa/workbook.py
  research-suite/app/charting/__init__.py
  research-suite/app/charting/disclosure.py
  research-suite/app/charting/export.py
  research-suite/app/charting/interlocks.py
  research-suite/app/charting/language.py
```

```

research-suite/app/charting/models.py
research-suite/app/charting/narrative.py
research-suite/app/charting/phi.py
research-suite/app/charting/proofing.py
research-suite/app/charting/routes.py
research-suite/app/charting/scales.py
research-suite/app/charting/specialty.py
research-suite/app/charting/store.py
research-suite/app/charting/systems.py
research-suite/app/compliance/__init__.py
research-suite/app/compliance/journals.py
research-suite/app/compliance/rubric.py
research-suite/app/compliance/simulator.py
research-suite/app/doctor.py
research-suite/app/evidence/__init__.py
research-suite/app/evidence/appraisal.py
research-suite/app/evidence/dedupe.py
research-suite/app/evidence/levels.py
research-suite/app/main.py
research-suite/app/models.py
research-suite/app/research/__init__.py
research-suite/app/research/pico.py
research-suite/app/security.py
research-suite/app/settings.py
research-suite/app/sources/__init__.py
research-suite/app/sources/base.py
research-suite/app/sources/fulltext.py
research-suite/app/sources/gov.py
research-suite/app/sources/importers.py
research-suite/app/sources/scholarly.py
research-suite/app/static/app.js
research-suite/app/static/chart.js
research-suite/app/static/index.html
research-suite/app/static/shell.js
research-suite/app/static/styles.css
research-suite/app/storage.py
research-suite/app/writing/__init__.py
research-suite/app/writing/draft.py
research-suite/app/writing/integrity.py
research-suite/app/writing/proof.py
research-suite/app/writing/statistics.py
research-suite/app/writing/style.py
research-suite/requirements.txt
research-suite/run.ps1
research-suite/run.sh
research-suite/tests/__init__.py
research-suite/tests/run_all.sh
research-suite/tests/test_apa_citations.py
research-suite/tests/test_charting_flow.py
research-suite/tests/test_charting_language.py
research-suite/tests/test_charting_proofing.py
research-suite/tests/test_charting_scales.py
research-suite/tests/test_deck.py
research-suite/tests/test_docx_layout.py
research-suite/tests/test_evidence.py
research-suite/tests/test_frontend.py
research-suite/tests/test_fulltext.py
research-suite/tests/test_research_tools.py
research-suite/tests/test_security.py
research-suite/tests/test_sources.py
== squash 750d52c: Add bonsai nursery tracker with scannable QR plant labels (#1)
.gitignore
README.md
css/styles.css
index.html

```

```

js/app.js
js/charts.js
js/qrcode.js
js/store.js
test/emit_matrices.js
test/smoke_app.py
test/verify_qr.py
== squash 9f8f639: Add environment factors, care profiles, target bands and optional cloud sync (#2)
README.md
css/styles.css
index.html
js/app.js
js/charts.js
js/profiles.js
js/store.js
js/sync.js
test/smoke_app.py
test/sync_test.py
== squash ab26e88: Align care profiles with the three handbooks; rename to Koch's Tree Nursery
Tracker (#3)
README.md
css/styles.css
index.html
js/app.js
js/profiles.js
js/store.js
test/smoke_app.py
== squash 3d97ca5: Add master calendar, derived drip log, and handbook task windows (#4)
README.md
css/styles.css
index.html
js/app.js
js/calendar.js
js/profiles.js
js/store.js
test/calendar_test.py
== squash 39ffeb7: Add weather watch, move alerts, and calendar export (#5)
README.md
css/styles.css
index.html
js/app.js
js/calendar.js
js/profiles.js
js/weather.js
test/smoke_app.py
test/weather_test.py
== squash 32ded12: Add professional record layer: specimen, lineage, IPM, photos, derived readouts
(#6)
README.md
css/styles.css
index.html
js/app.js
js/photos.js
js/profiles.js
js/store.js
test/pro_test.py
test/smoke_app.py

```